

Dr. Sajid Muhaimin Choudhury

Associate Professor, Department of EEE, BUET

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Brief Biography

Dr. Sajid Choudhury is an Associate Professor in the Department of EEE, BUET. He completed his Ph.D. from Purdue University, IN, USA. His current research interest is in Photonic Quantum Computing, Photovoltaic Cells, Flat Optics with Metasurface, Photonic Devices with Phase Change Materials, Embedded Systems Design. Dr. Choudhury is a senior member of the IEEE, immediate-past-chair, IEEE Photonics Society Bangladesh Chapter, founding President, The Optica Bangladesh Section. He is a member of the National Young Academy of Bangladesh (NYAB).

Experience

Associate Professor, Department of Electrical and Electronic Engineering (EEE) Bangladesh University of Engineering and Technology (BUET)	July 2022 – to date
Assistant Professor, Department of Electrical and Electronic Engineering (EEE) Bangladesh University of Engineering and Technology (BUET)	June 2013 – July 2022
Lecturer, Department of Electrical and Electronic Engineering (EEE) Bangladesh University of Engineering and Technology (BUET)	Jan 2010 – June 2013
Lecturer, Institute of Information and Communication Technology (IICT) Bangladesh University of Engineering and Technology (BUET)	Nov 2009 – Jan 2010

Education

Ph.D.	Purdue University , West Lafayette, IN, USA School of Electrical and Computer Engineering Ph.D. Thesis: WAVEFRONT MANIPULATION WITH META-SURFACES BASED ON NEW MATERIALS Ph.D. Co-supervisor(s): Alexandra Boltasseva and Alexander Kildishev	Aug 2013 – Aug 2019
M.Sc.	Bangladesh University of Engineering and Technology (BUET) Department of Electrical and Electronic Engineering M.Sc. Engg. Thesis: Design of a Fractal Antenna based on Hexaflake Fractal Structure M.Sc. Engg. Supervisor: Dr. M. A. Matin	Aug 2011 – 2013
B.Sc.	Bangladesh University of Engineering and Technology (BUET) Department of Electrical and Electronic Engineering - CGPA: 3.94/4.0 Undergraduate Thesis: Design and Analysis of a Multiband Dual Feed Axially Symmetric Cassegrain Antenna System Undergraduate Supervisor: Dr. M. A. Matin	Dec 2004 – Aug 2010
H.S.C.	Notre Dame College , Dhaka GPA: 5.00/5.00	2004
S.S.C.	Udayan Uchchya Madhyamic Bidyalaya , Dhaka GPA: 5.00/5.00	2002

Details of Publications from 20/05/2024 to 21/05/2025 _____

Details of Publications since last appointment _____

Details of Important design/research projects from 20/05/2024 to 21/05/2025 _____

Postgraduate thesis supervised _____

- Md Asif Hossain Bhuiyan, M.Sc. Engg. (May 2025) **Polarization Insensitive Electrically Reconfigurable Metasurface For Metalensing At Near Infrared Waveband**
- Md. Mahfuzul Haque, M.Sc. Engg. (Jan 2025) **Design Of Silicon-carbide Based Single-quantum-well White LED**
- Md. Ehsanul Karim, M.Sc. Engg. (June 2024) **Phase Change Material Based Broadband Multifunctional Metasurface for the Visible Range**
- Shamima Akter Mitu M.Sc. Engg. (May 2023) **Design of an All-optical Plasmonic Modulator for Two Micrometer Waveband**

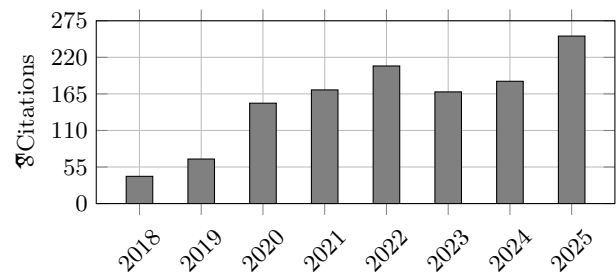
Complete List of Publications _____

Publications Metrics

(a) Google Scholar Metrics

Total Citations	1308
h-index	14
i10-index	12

(b) Google Scholar Citations per Year



Journal Articles

- [J27] Md. Mahfuzul Haque and Sajid Muhaimin Choudhury. “Tailoring electronic and optoelectronic properties of 2D-SiC via defects and doping: a first-principles study toward efficient white light-emitting diodes”. In: *RSC advances* 15 (2025), pp. 29335–29366. 10.1039/D5RA04586J [🔗](#). Citations: 1, SJR Q1(IF 3.9).
- [J27] Md. Mahfuzul Haque, Md. Rasidul Islam, and Sajid Muhaimin Choudhury. “Investigation of the physical properties through strain effect of monolayer silicon carbide material: DFT analysis”. In: *Physica B: Condensed Matter* 697 (2025), p. 416670. 10.1016/j.physb.2024.416670 [🔗](#). Citations: 3, SJR Q2(IF 2.4).
- [J27] Md. Ehsanul Karim, Md. Redwanul Karim, and Sajid Muhaimin Choudhury. “Synergizing deep learning and phase change materials for four-state broadband multifunctional metasurfaces in the visible range”. In: *Optics & Laser Technology* 181 (2025), p. 111730. 10.1016/j.optlastec.2024.111730 [🔗](#). Citations: 5, SJR Q1(IF 5.2).
- [J27] Ayon Sarker and Sajid Muhaimin Choudhury. “Concentric annular-hexagonal plasmonic resonator with nanorod vertices for dual-band absorption in NIR and MIR for sensing applications”. In: *Opt. Continuum* 4.5 (May 2025), pp. 1159–1173. 10.1364/OPTCON.558501 [🔗](#). URL: <https://opg.optica.org/optcon/abstract.cfm?URI=optcon-4-5-1159>. SJR Q3(IF 1.1).

- [J27] Md. Asif Hossain Bhuiyan, Purbayan Das, and Sajid Muhaimin Choudhury. “Polarization insensitive electrically reconfigurable meta-lens for the 2 μ m wavelength”. In: *Opt. Mater. Express* 14.12 (Dec. 2024), pp. 2830–2843. 10.1364/OME.540435 [✉](#). [SJRR Q2\(IF 2.8\)](#).
- [J27] Md. Ehsanul Karim and Sajid Muhaimin Choudhury. “Sb2S3/AlGaAs based Reconfigurable Metasurface for Dynamic Polarization and Directionality Control of Quantum Emitter Emission”. In: *RSC Advances* 40 (2024). 10.1039/D4RA03726J [✉](#). [Citations: 1, SJRR Q1\(IF 3.9\)](#).
- [J27] Prithu Mahmud, Kaniz Fatema Supti, and Sajid Muhaimin Choudhury. “Lithium niobate photonic topological insulator-based multi-wavelength optical demultiplexer with piezoelectric switch-off”. In: *Opt. Express* 32.26 (Dec. 2024), pp. 45786–45800. 10.1364/OE.541271 [✉](#). [Citations: 1, SJRR Q1\(IF 3.2\)](#).
- [J27] Mehedi Hasan Himel, Bejoy Sikder, Tanvir Ahmed, and Sajid Muhaimin Choudhury. “Biomimicry in Nanotechnology: A Comprehensive Review”. In: *NanoScale Advances* 5 (2023), pp. 595–614. 10.1039/D2NA00571A [✉](#). [Citations: 36, SJRR Q1\(IF 5.6\)](#).
- [J27] Md. Ehsanul Karim and Sajid Muhaimin Choudhury. “Reconfigurable Broadband Metasurface with Switchable Functionalities in the Visible Range”. In: *Optical Materials Express* 13.5 (2023), pp. 1409–1423. 10.1364/OME.489981 [✉](#). [Citations: 7, SJRR Q2\(IF 2.8\)](#).
- [J27] Md Asif Hossain Bhuiyan, Shamima Akter Mitu, and Sajid Muhaimin Choudhury. “TiN-GST-TiN all-optical reflection modulator for the 2 μ m wave band reaching 85% efficiency”. In: *Applied Optics* 61 (2022), pp. 9262–9270. 10.1364/AO.470247 [✉](#). [Citations: 3, SJRR Q2\(IF 2.08\)](#).
- [J27] Himaddri Roy, Ehsanul Karim, Sujoy Mondal, and Sajid Muhaimin Choudhury. “Custom Gold-Patterned Rewritable Optical Disc based Plasmonic Sensor for Blood Hemoglobin Detection”. In: *Optics Continuum* 1.10 (2022). 10.1364/OPTCON.473106 [✉](#). [Citations: 1, SJRR Q3\(IF 1.1\)](#).
- [J27] Yasir Fatha Abed, Md Asif Hossain Bhuiyan, and Sajid Muhaimin Choudhury. “T Grating on Nano-Cavity Array based Refractive Index Sensor”. In: *J. Opt. Soc. Am. B* 39.9 (2021). 10.1364/JOSAB.426526 [✉](#). [Citations: 1, SJRR Q2\(IF 2.18\)](#).
- [J27] Mohammad Muntasir Hassan, Farhan Sadik Sium, Fariba Islam, and Sajid Muhaimin Choudhury. “A Review on Plasmonic Nano-biosensors for Virus Detection with a Focus on Coronavirus”. In: *Sensing and Bio-Sensing Research* 33 (2021), p. 100429. 10.1016/j.sbsr.2021.100429 [✉](#). [Citations: 82, SJRR Q2\(IF 4.42\)](#).
- [J27] Ayon Sarker, Shamima Akter Mitu, Purbayan Das, and Sajid Muhaimin Choudhury. “Structurally Tunable Gear-Shaped Plasmonic Sensor”. In: *Optics Express* 28.24 (2020), pp. 36070–36083. 10.1364/OE.410123 [✉](#). [Citations: 14, SJRR Q1\(IF 3.67\)](#).
- [J27] Huan Jiang, Sajid Muhaimin Choudhury, Zhaxylyk A Kudyshev, Di Wang, Peng Xiao, Yongyuan Jiang, and Alexander V Kildishev. “Enhancing sensitivity to ambient refractive index with tunable few-layer graphene/hBN nanoribbons”. In: *Photonics Research* 7.7 (2019), pp. 815–822. 10.1364/PRJ.7.000815 [✉](#). [Citations: 36, SJRR Q1\(IF 7.37\)](#).
- [J27] Huan Jiang, Harsha Reddy, Deesha Shah, Zhaxylyk A Kudyshev, Sajid Muhaimin Choudhury, Di Wang, Yongyuan Jiang, and Alexander V Kildishev. “Modulating Phase by Metasurfaces with Gated Ultra-thin TiN Films”. In: *Nanoscale* 11 (2019), pp. 11167–11172. 10.1039/C9NR00205G [✉](#). [Citations: 10, SJRR Q1\(IF 6.62\)](#).
- [J27] Oscar Quevedo-Teruel, Hongsheng Chen, Ana Díaz-Rubio, Gurkan Gok, Anthony Grbic, Gabriele Minatti, Enrica Martini, Stefano Maci, George V Eleftheriades, Michael Chen, Nikolay I Zheludev, Nikitas Papasimakis, Sajid Muhaimin Choudhury, Zhaxylyk A Kudyshev, Soham Saha, Harsha Reddy, Alexandra Boltasseva, Vladimir M Shalaev, Alexander V Kildishev, Daniel Sievenpiper, Christophe Caloz, Andrea Alù, Qiong He, Lei Zhou, Guido Valerio, Eva Rajo-Iglesias, Zvonimir Sipus, Francisco Mesa, Raul Rodríguez-Berral, Francisco Medina, Victor Asadchy, Sergei Tretyakov, and Christophe Craeye. “Roadmap on metasurfaces”. In: *Journal of Optics* 21.7 (2019), p. 073002. 10.1088/2040-8986/ab161d [✉](#). [Citations: 306, SJRR Q2\(IF 2.66\)](#).
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- [J27] Vladimir A. Zenin, Sajid Muhaimin Choudhury, Soham Saha, Vladimir M. Shalaev, Alexandra Boltasseva, and Sergey I. Bozhevolnyi. “Hybrid plasmonic waveguides formed by metal coating of dielectric ridges”. In: *Optics Express* 25.11 (2017), pp. 12295–12302. 10.1364/OE.25.012295 . Citations: 38, SJR Q1(IF 3.67).
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- [J27] MA Zaman, SA Mamun, Md Gaffar, SM Choudhury, Md M Alam, and Md Matin. “Phased Array Synthesis Using Modified Particle Swarm Optimization”. In: *Journal of Engineering Science & Technology Review* 4.1 (2011). 10.25103/jestr.041.10 . Citations: 19, SJR Q3(IF 1.6).
- [J27] Mohammad Asif Zaman, Md Gaffar, Md Mushfiqul Alam, Sayed Ashraf Mamun, Sajid Muhaimin Choudhury, and MA Matin. “Approximate Closed-Form Expression of the Electric Field of a Conical Horn Antenna”. In: *International Journal of Computer and Electrical Engineering* 3.1 (2011), p. 48. URL: <http://ijcee.org/papers/291-E337.pdf>. Citations: 1,

Conference Proceedings

- [C28] Sajid Muhaimin Choudhury, Md. Ehsanul Karim, Md. Redwanul Karim, and Naved Sadat Yamin. “Integrating Deep Learning and Topology Optimization for Next-Generation Nanophotonic Devices”. In: *2025 Photonics & Electromagnetics Research Symposium - Spring (PIERS-Spring)*. 2025. 10.1109/PIERS-Spring66516.2025.11276753 .
- [C28] Md. Mahfuzul Haque and Sajid Muhaimin Choudhury. “Towards High-Performance Quantum-Based LEDs: Optoelectronic Modulation of 2D Silicon Carbide for Tunable and Efficient White Light Emission”. In: *2025 Photonics & Electromagnetics Research Symposium - Spring (PIERS-Spring)*. 2025. 10.1109/PIERS-Spring66516.2025.11276753 .
- [C28] Md Al Shahriar Shakil and Sajid Muhaimin Choudhury. “DFT Analysis of Strain Effect in Bilayer 2D-SiC on Electrical, Optical, Phonon, and Thermodynamic Properties”. In: *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*. 2025. 10.1109/QPAIN66474.2025.11172078 .
- [C28] Naved Sadat Yamin and Sajid Muhaimin Choudhury. “Valley Photonic Topological Insulator for Fluorescence Endoscopy”. In: *2025 Photonics & Electromagnetics Research Symposium - Spring (PIERS-Spring)*. 2025. 10.1109/PIERS-Spring66516.2025.11276753 .
- [C28] Mohammad Tahsin Alam, Yasir Mahmud, Zafrin Jahan Nikita, and Sajid Muhaimin Choudhury. “Gesture Controlled Bot with Temperature & Humidity (TH) Sensing Features”. In: *2024 2nd International Conference on Information and Communication Technology (ICICT)*. 2024, pp. 36–40. 10.1109/ICICT64387.2024.10839649 .
- [C28] Md Mahfuzul Haque and Sajid Muhaimin Choudhury. “DFT Analysis of Strain Effect in Bilayer 2D-SiC on Electrical, Optical, Phonon, and Thermodynamic Properties”. In: *2024 13th International Conference on Electrical and Computer Engineering (ICECE)*. 2024. 10.1109/ICECE64886.2024.11024753  Citations: 1,
- [C28] Md Asif Hossain Bhuiyan, Shamima Akter Mitu, and Sajid Muhaimin Choudhury. “VO₂-based All-optical Reflection Modulator for 2 μ m Wave Band”. In: *2023 IEEE Photonics Conference (IPC)*. 2023, pp. 1–2. 10.1109/IPC57732.2023.10360477  Citations: 1,

- [C28] Saleh Ahmed Khan, Sadat Tahmeed Azad, Tiasa Mondal, Abdullah Jubair Bin Iqbal, and Sajid Muhaimin Choudhury. "Development of an Internet of Things based Bangla Calendar Clock". In: *2023 26th International Conference on Computer and Information Technology (ICCIT)*. 2023, pp. 1–6. 10.1109/ICCIT60459.2023.10441436 [↗](#).
- [C28] Abdul Mukit, Md Sabbir Hossen Bijoy, Sajid Muhaimin Choudhury, and Md Tareq Mahmud. "Discrete Modulated Continuous-Variable Quantum Key Distribution: Security and Noise Tolerance Enhanced by Decoy States and Effective Error Correction Protocol Integration". In: *2023 IEEE International Conference on Telecommunications and Photonics (ICTP)*. 2023, pp. 1–5. 10.1109/ICTP60248.2023.10490525 [↗](#) Citations: 2,
- [C28] Kushol Roy Pritom, Md. Ehsanul Karim, and Sajid Muhaimin Choudhury. "A Polarization Insensitive Achromatic Metalens Operating at Two Wavelengths in Visible Regime". In: *2023 IEEE International Conference on Telecommunications and Photonics (ICTP)*. 2023, pp. 01–05. 10.1109/ICTP60248.2023.10491019 [↗](#) Citations: 1,
- [C28] Soikot Sarkar and Sajid Muhaimin Choudhury. "Design and Performance Analysis of a c-Si Thin-Film Solar Cell Using Plasmonic Ag Nanostructures". In: *2023 IEEE International Conference on Telecommunications and Photonics (ICTP)*. 2023, pp. 01–05. 10.1109/ICTP60248.2023.10490886 [↗](#).
- [C28] Zhaxylyk A. Kudyshev, Ludmila J. Prokopeva, Maowen Song, Sajid Muhaimin Choudhury, and Alexander V. Kildishev. "Bi-anisotropic homogenization for efficient metasurface design (invited)". In: *2018 International Applied Computational Electromagnetics Society Symposium (ACES)*. 2018, pp. 1–2. 10.23919/ROPACES.2018.8364134 [↗](#).
- [C28] Sajid Muhaimin Choudhury, Vladimir A. Zenin, Soham Saha, Vladimir M. Shalae, Sergei Bozhevolnyi, and Alexandra Boltasseva. "Novel Hard Mask Fabrication Method for Hybrid Plasmonic Waveguide and Metasurfaces". In: *Frontiers in Optics 2017*. Optica Publishing Group, 2017, JTu2A.12. 10.1364/FIO.2017.JTu2A.12 [↗](#).
- [C28] Sajid Muhaimin Choudhury, Amr Shaltout, Vladimir M. Shalae, Alexander V. Kildishev, and Alexandra Boltasseva. "Experimental Realization of Color Hologram Using Pancharatnam-Berry Phase Manipulating Metasurface". In: *Conference on Lasers and Electro-Optics*. Optica Publishing Group, 2016, FF1D.8. 10.1364/CLEO_QELS.2016.FF1D.8 [↗](#) Citations: 2,
- [C28] Sajid Muhaimin Choudhury, Amr Shaltout, Vladimir M. Shalae, Alexandra Boltasseva, and Alexander V. Kildishev. "Color Hologram Generation Using a Pancharatnam-Berry Phase Manipulating Metasurface". In: *CLEO: 2015*. Optica Publishing Group, 2015, JTu5A.89. 10.1364/CLEO_AT.2015.JTu5A.89 [↗](#) Citations: 4,
- [C28] Parvez Ahmmed, Zabir Ahmed, Mohammad Ishfaq Jahan Rafee, M A Awal, and Sajid Muhaimin Choudhury. "Self-localization of a mobile robot using monocular vision of a chessboard pattern". In: *8th International Conference on Electrical and Computer Engineering*. 2014, pp. 753–756. 10.1109/ICECE.2014.7026828 [↗](#) Citations: 7,
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- [C28] S. M. L. Kabir, M. S. Hussain, S. M. Choudhury, and A. H. Chowdhury. "Developing A Low-Cost Multiple Motor Switched Photovoltaic Powered Irrigation System". In: *Proceedings of the 3rd International Conference on Water and Flood Management (ICWFM-2011)*. Vol. 2. 2011, pp. 577–581.
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- [C28] Md. Gaffar, Sajid Muhaimin Choudhury, Mohammad Asif Zaman, Md. Imran Momtaz, M. Shah Alam, and Md. Abdul Matin. “Sensitivity analysis of a circularly polarized U-slot microstrip antenna”. In: *International Conference on Electrical & Computer Engineering (ICECE 2010)*. 2010, pp. 546–549. 10.1109/ICECE.2010.5700750  Citations: 2,
- [C28] Mohammad Asif Zaman, Md. Gaffar, Sajid Muhaimin Choudhury, and M. A. Matin. “Optimization and analysis of a Ka band Pickett Potter horn antenna with low cross polarization”. In: *International Conference on Electrical & Computer Engineering (ICECE 2010)*. 2010, pp. 542–545. 10.1109/ICECE.2010.5700749  Citations: 12,
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- [C28] Sajid Muhaimin Choudhury. “Design and implementation of a low cost Power Factor Improvement device”. In: *TENCON 2008 - 2008 IEEE Region 10 Conference*. 2008, pp. 1–4. 10.1109/TENCON.2008.4766529  Citations: 21,

Patents

- [P3] Esteban Marinero-Caceres, Arnold Toppo, Sajid Choudhury, Urcan Guler, Zhaxylyk Kudyshev, Joseph Pekny, Swati Pol, Harsha Reddy, and Vladimir Shalaev. “Thermophotovoltaic system and method of making the same”. US20210234498A1. July 29, 2021. URL: <https://patents.google.com/patent/US20210234498A1/en>.
- [P3] Amr Shaltout, Sajid Choudhury, Alexander V. Kildishev, Alexandra Boltasseva, and Vladimir M. Shalaev. “System for producing ultra-thin color phase hologram with metasurfaces”. US9952557B2. Apr. 24, 2018. URL: <https://patents.google.com/patent/US9952557B2/en> Citations: 13,

Preprint / Manuscript Under Preparation

- [X5] Subhan Zawad Bihan, Anindya Kishore Choudhury, and Sajid Muhaimin Choudhury. “Exploiting Steane-Code Syndrome Redundancy in Hybrid Photonic-Stabilizer One-Way Repeaters”. manuscript under preparation. 2026.
- [X5] Puja Das, Md. Kawsar Alam, and Sajid Muhaimin Choudhury. “HBT Perovskite/CdS Solar Cell Reaching Efficiency Over 35% Using Periodic Triangular Gratings”. manuscript under preparation. 2026.
- [X5] Junayet Hossain, Sajid Muhaimin Choudhury, and Mohammed Imamul Hasan Bhuiyan. “Harnessing Topological Valley-Hall States in Photonic Crystals for Robust Cancer Cell Detection”. ArXiv. 2025. URL: <https://arxiv.org/abs/2509.05690>.
- [X5] Dipika Rani Nath, Sadid Muneer, and Sajid Muhaimin Choudhury. “All-Optical Varifocal Switching in a Polarization-Insensitive Si-GST Metalens”. ArXiv. 2025. URL: <https://arxiv.org/abs/2511.21138>.

Membership / Fellowship of Learned Societies, Professional Institutions and Other Noteworthy Affiliations ---

Senior Member, Institute of Electrical and Electronic Engineers (IEEE)

- Secretary, IEEE Bangladesh Section (July 2021 - May 2022)
- Chair, IEEE Young Professionals Bangladesh (Mar 2020 - Apr 2022)
- Chair, IEEE Graduates of the Last Decade (Jan 2013 - Dec 2013)
- Vice -Chair, IEEE Graduates of the Last Decade (Jan 2011 - Dec 2012)
- Student Activities Coordinator, IEEE Bangladesh Section (Jan 2011 - Dec 2012)
- Chair, IEEE BUET Student Branch (Jan 2008 - Aug 2009)
- Treasurer, IEEE BUET Student Branch (Jan 2007 - Dec 2008)

Member, IEEE Photonics Society

- Vice Chair, IEEE Photonics Society Bangladesh (April 2022 – to date)
- Founding Chair, IEEE Photonics Society Bangladesh (Mar 2021 – Apr 2022)

Member, The Optica

- Founding President, Optica Bangladesh Section May 2022 – to date
- Founding Moderator, BUET Optical and Photonics Society July 2022 – to date
- Treasurer, OSA Purdue Chapter, USA Jun 2016 – May 2017

Member, National Young Academy of Bangladesh (NYAB), April 2021 – to date

Life Member, American Alumni Association of Bangladesh (AAAB) , April 2024 – to date

Life Member, Association of BUET Alumni, April 2021 – to date

Student Activities at Purdue University, West Lafayette, IN, USA

- President, **Nanotechnology Student Advisory Council (NSAC)** (Jun 2017 – May 2018)
- Vice-President, **Nanotechnology Student Advisory Council (NSAC)** (Jun 2016 – May 2017)
- Treasurer, SPIE Purdue Chapter, USA (Jun 2015 – May 2016)
- President, Bangladesh Students Association (**Purdue BDSA**), USA (Jul 2017 – Jun 2018)
- Treasurer, Bangladesh Students Association (**Purdue BDSA**), USA (Jul 2015 – Jun 2017)

References ---

Available upon request